Functional Specifications of Standard API Interface

- **Company:** ZhongNeng EV
- **Version:** 1.0
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Document Version

Version	Modifications	Date	Author
1.0	Initial Draft	2026-04-24	Backend Team

Detailed API Specification

1. User Registration

Register a new customer account in the EV Seminar Registration System. After successful registration, an OTP verification email will be sent to the provided email address.

- **Url [POST]:** `/api/auth/register`
- **Input Parameters:**
  - **Header:**
    - Accept: `application/json`
    - Content-Type: `application/json`
  - **Path Variables / Query Parameters:**
    - None
  - **Body:**

Name	Mandatory/ Optional	Data Type	Validation	Sample / Possible Values	Remark
name	M	String	Max 100 characters, Not blank	"John Doe"	Customer full name
telephone	M	String	Max 20 characters, Not blank	"+852 9123 4567"	Contact telephone number
email	M	String	Valid email format, Not blank	"john.doe@example.com"	Must be unique in system
password	M	String	Min 6 characters, Not blank	"password123"	Will be encrypted

Name	Mandatory/ Optional	Data Type	Validation	Sample / Possible Values	Remark
					before storage
customerType	M	Enum	Must be COMPANY or PERSONAL	"PERSONAL"	Customer account type

- **Output in JSON format:**

Response Sample (Success)

```
{
    "message": "Registration successful. Please check your email for verification code."
}
```

Response Sample (Fail - Validation Error)

```
{
    "status": 400,
    "message": "Validation failed",
    "errors": [
        "email: Invalid email format",
        "password: Password must be at least 6 characters"
    ],
    "timestamp": "2026-04-24T19:30:00"
}
```

Response Sample (Fail - Duplicate Email)

```
{
    "status": 400,
    "message": "Email already registered",
    "errors": ["Email already registered"],
    "timestamp": "2026-04-24T19:30:00"
}
```

- **Rules and Validations:**
  - Email must be unique in the system
  - Password must be at least 6 characters long
  - Customer type must be either COMPANY or PERSONAL
  - Upon successful registration, a 6-digit OTP code is generated and sent to the email
  - Account remains unverified until email verification is completed

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2. Email Verification

Verify customer email address using the OTP code sent during registration. Upon successful verification, a JWT token is returned for authentication.

- **Url [POST]:** `/api/auth/verify-email`
- **Input Parameters:**
  - **Header:**
    - Accept: `application/json`
    - Content-Type: `application/json`
  - **Path Variables / Query Parameters:**
    - None
  - **Body:**

Name	Mandatory/ Optional	Data Type	Validation	Sample / Possible Values	Remark
email	M	String	Valid email format, Not blank	"john.doe@example.com"	Registered email address
otpCode	M	String	Exactly 6 characters, Not blank	"123456"	OTP code received via email

- **Output in JSON format:**

Response Sample (Success)

```
{
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "email": "john.doe@example.com",
    "name": "John Doe",
    "verified": true
}
```

Response Sample (Fail - Invalid OTP)

```
{
    "status": 400,
    "message": "Invalid or expired OTP code",
    "errors": ["Invalid or expired OTP code"],
    "timestamp": "2026-04-24T19:30:00"
}
```

Response Sample (Fail - User Not Found)

```
{
    "status": 404,
    "message": "User not found",
    "errors": ["User not found"],
    "timestamp": "2026-04-24T19:30:00"
}
```

- **Rules and Validations:**
  - OTP code must be exactly 6 digits
  - OTP code expires after **10 minutes** from the time of registration
  - Email must match a registered but unverified account
  - Upon successful verification, emailVerified flag is set to true
  - JWT token is generated and returned for subsequent authenticated requests

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3. User Login

Authenticate an existing customer and obtain a JWT token for accessing protected endpoints.

- **Url [POST]:** `/api/auth/login`
- **Input Parameters:**
  - **Header:**
    - Accept: `application/json`
    - Content-Type: `application/json`
  - **Path Variables / Query Parameters:**
    - None
  - **Body:**

Name	Mandatory/ Optional	Data Type	Validation	Sample / Possible Values	Remark
email	M	String	Valid email format, Not blank	"john.doe@example.com"	Registered email address
password	M	String	Not blank	"password123"	Account password

- **Output in JSON format:**

Response Sample (Success)

```
{
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "email": "john.doe@example.com",
    "name": "John Doe",
```

```
    "verified": true
}
```

Response Sample (Fail - Invalid Credentials)

```
{
    "status": 401,
    "message": "Invalid email or password",
    "errors": ["Invalid email or password"],
    "timestamp": "2026-04-24T19:30:00"
}
```

Response Sample (Fail - Email Not Verified)

```
{
    "status": 401,
    "message": "Email not verified. Please verify your email first.",
    "errors": ["Email not verified. Please verify your email first."],
    "timestamp": "2026-04-24T19:30:00"
}
```

- **Rules and Validations:**
  - Email and password must match an existing account
  - Account must be email-verified before login is allowed
  - Password is compared against encrypted stored password
  - JWT token is generated upon successful authentication
  - JWT token expires after **24 hours** (configured as `86400000 ms` in `application.yml`)
  - No refresh token mechanism is implemented
  - Token should be included in Authorization header for subsequent requests

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4. Get Customer Profile

Retrieve the authenticated customer's profile information.

- **Url [GET]:** `/api/users/profile`
- **Input Parameters:**
  - **Header:**
    - Accept: `application/json`
    - Authorization: `Bearer {JWT_TOKEN}` (Required)
  - **Path Variables / Query Parameters:**
    - None
  - **Body:**
    - None

- **Output in JSON format:**

Response Sample (Success)

```
{
    "id": 1,
    "name": "John Doe",
    "telephone": "+852 9123 4567",
    "email": "john.doe@example.com",
    "customerType": "PERSONAL",
    "emailVerified": true,
    "createdAt": "2026-04-20T10:30:00"
}
```

Response Sample (Fail - Unauthorized)

```
{
    "status": 401,
    "message": "Unauthorized access",
    "errors": ["Unauthorized access"],
    "timestamp": "2026-04-24T19:30:00"
}
```

Response Sample (Fail - Invalid Token)

```
{
    "status": 401,
    "message": "Invalid or expired token",
    "errors": ["Invalid or expired token"],
    "timestamp": "2026-04-24T19:30:00"
}
```

- **Rules and Validations:**
  - Valid JWT token must be provided in Authorization header
  - Token must not be expired
  - User extracted from token must exist in the system
  - Returns profile information for the authenticated user only

---

5. Get All Vehicles

Retrieve a list of all available electric vehicles in the catalog.

- **Url [GET]:** `/api/vehicles`

- **Input Parameters:**

- **Header:**
  - Accept: `application/json`
- **Path Variables / Query Parameters:**
  - None
- **Body:**
  - None

- **Output in JSON format:**

Response Sample (Success)

```
[
    {
        "id": 1,
        "modelNumber": "EV-2024-001",
        "description": "Tesla Model 3 - Long Range AWD",
        "pictureUrl": "https://example.com/images/tesla-model3.jpg",
        "features": "Autopilot, Premium Interior, 358 miles range",
        "unitPrice": 45000.00
    },
    {
        "id": 2,
        "modelNumber": "EV-2024-002",
        "description": "BYD Seal - Premium Edition",
        "pictureUrl": "https://example.com/images/byd-seal.jpg",
        "features": "Advanced Driver Assistance, Luxury Interior, 520km range",
        "unitPrice": 38000.00
    }
]
```

Response Sample (Fail - Server Error)

```
{
    "status": 500,
    "message": "An unexpected error occurred",
    "errors": ["An unexpected error occurred"],
    "timestamp": "2026-04-24T19:30:00"
}
```

- **Rules and Validations:**
  - No authentication required (public endpoint)
  - Returns all vehicles in the database
  - Empty array returned if no vehicles exist
  - Unit price is in decimal format with 2 decimal places

---

6. Get Vehicle by ID

Retrieve detailed information about a specific electric vehicle.

- **Url [GET]:** `/api/vehicles/{id}`
- **Input Parameters:**
  - **Header:**
    - Accept: `application/json`
  - **Path Variables / Query Parameters:**
    - `id`: Vehicle ID (Long, Required)
  - **Body:**
    - None
- **Output in JSON format:**

Response Sample (Success)

```
{
    "id": 1,
    "modelNumber": "EV-2024-001",
    "description": "Tesla Model 3 - Long Range AWD",
    "pictureUrl": "https://example.com/images/tesla-model3.jpg",
    "features": "Autopilot, Premium Interior, 358 miles range",
    "unitPrice": 45000.00
}
```

Response Sample (Fail - Not Found)

```
{
    "status": 404,
    "message": "Vehicle not found with id: 999",
    "errors": ["Vehicle not found with id: 999"],
    "timestamp": "2026-04-24T19:30:00"
}
```

- **Rules and Validations:**
  - No authentication required (public endpoint)
  - Vehicle ID must exist in the database
  - Returns 404 if vehicle with specified ID is not found

---

7. Get Upcoming Seminars

Retrieve a list of all upcoming seminars with available seats.

- **Url [GET]:** `/api/seminars/upcoming`
- **Input Parameters:**

- **Header:**
  - Accept: `application/json`
- **Path Variables / Query Parameters:**
  - None
- **Body:**
  - None

- **Output in JSON format:**

Response Sample (Success)

```
[
    {
        "id": 1,
        "vehicleId": 1,
        "vehicleModelNumber": "EV-2024-001",
        "vehicleDescription": "Tesla Model 3 - Long Range AWD",
        "seminarDate": "2026-05-15T14:00:00",
        "maxSeats": 50,
        "bookedSeats": 35,
        "availableSeats": 15
    },
    {
        "id": 2,
        "vehicleId": 2,
        "vehicleModelNumber": "EV-2024-002",
        "vehicleDescription": "BYD Seal - Premium Edition",
        "seminarDate": "2026-05-20T10:00:00",
        "maxSeats": 40,
        "bookedSeats": 40,
        "availableSeats": 0
    }
]
```

Response Sample (Fail - Server Error)

```
{
    "status": 500,
    "message": "An unexpected error occurred",
    "errors": ["An unexpected error occurred"],
    "timestamp": "2026-04-24T19:30:00"
}
```

- **Rules and Validations:**
  - No authentication required (public endpoint)
  - Returns only seminars scheduled for future dates
  - Available seats calculated as: maxSeats - bookedSeats
  - Seminars with 0 available seats are still included in the list

- Empty array returned if no upcoming seminars exist

---

8. Register for Seminar

Register the authenticated customer for a specific seminar with a specified number of seats.

- **Url [POST]:** `/api/seminars/{seminarId}/register`
- **Input Parameters:**
  - **Header:**
    - Accept: `application/json`
    - Content-Type: `application/json`
    - Authorization: `Bearer {JWT_TOKEN}` (Required)
  - **Path Variables / Query Parameters:**
    - `seminarId`: Seminar ID (Long, Required)
  - **Body:**

Name	Mandatory/ Optional	Data Type	Validation	Sample / Possible Values	Remark
seatsBooked	M	Integer	Min: 1, Max: 2	1 or 2	Number of seats to book

- **Output in JSON format:**

Response Sample (Success - Confirmed)

```
{
    "id": 101,
    "seminarId": 1,
    "vehicleModelNumber": "EV-2024-001",
    "seminarDate": "2026-05-15T14:00:00",
    "seatsBooked": 2,
    "status": "SUCCESS",
    "createdAt": "2026-04-24T19:30:00",
    "updatedAt": "2026-04-24T19:30:00"
}
```

Response Sample (Success - Waitlisted)

```
{
    "id": 102,
    "seminarId": 1,
    "vehicleModelNumber": "EV-2024-001",
    "seminarDate": "2026-05-15T14:00:00",
    "seatsBooked": 2,
    "status": "WAIT",
    "createdAt": "2026-04-24T19:30:00",
```

```
    "updatedAt": "2026-04-24T19:30:00"
}
```

Response Sample (Fail - Validation Error)

```
{
    "status": 400,
    "message": "Validation failed",
    "errors": [
        "seatsBooked: Maximum 2 seats per registration"
    ],
    "timestamp": "2026-04-24T19:30:00"
}
```

Response Sample (Fail - Seminar Not Found)

```
{
    "status": 404,
    "message": "Seminar not found with id: 999",
    "errors": ["Seminar not found with id: 999"],
    "timestamp": "2026-04-24T19:30:00"
}
```

Response Sample (Fail - Unauthorized)

```
{
    "status": 401,
    "message": "Unauthorized access",
    "errors": ["Unauthorized access"],
    "timestamp": "2026-04-24T19:30:00"
}
```

- **Rules and Validations:**
  - Valid JWT token required in Authorization header
  - Seats booked must be between 1 and 2 (inclusive)
  - Seminar must exist and be scheduled for a future date
  - If sufficient seats available: status = SUCCESS
  - If insufficient seats available: status = WAIT (waitlist)
  - A customer with an active registration (status = SUCCESS or WAIT) cannot register for the same seminar again — returns HTTP 400 with message: `"You already have an active registration for this seminar"`
  - Registration status can be SUCCESS, WAIT, or CANCEL

---

9. Cancel Registration

Cancel an existing seminar registration for the authenticated customer.

- **Url [PUT]:** `/api/registrations/{id}/cancel`
- **Input Parameters:**
  - **Header:**
    - Accept: `application/json`
    - Authorization: `Bearer {JWT_TOKEN}` (Required)
  - **Path Variables / Query Parameters:**
    - `id`: Registration ID (Long, Required)
  - **Body:**
    - None
- **Output in JSON format:**

Response Sample (Success)

```
{
    "id": 101,
    "seminarId": 1,
    "vehicleModelNumber": "EV-2024-001",
    "seminarDate": "2026-05-15T14:00:00",
    "seatsBooked": 2,
    "status": "CANCEL",
    "createdAt": "2026-04-24T19:30:00",
    "updatedAt": "2026-04-24T20:00:00"
}
```

Response Sample (Fail - Registration Not Found)

```
{
    "status": 404,
    "message": "Registration not found with id: 999",
    "errors": ["Registration not found with id: 999"],
    "timestamp": "2026-04-24T19:30:00"
}
```

Response Sample (Fail - Unauthorized)

```
{
    "status": 401,
    "message": "You are not authorized to cancel this registration",
    "errors": ["You are not authorized to cancel this registration"],
    "timestamp": "2026-04-24T19:30:00"
}
```

Response Sample (Fail - Already Cancelled)

```
{
    "status": 400,
    "message": "Registration is already cancelled",
    "errors": ["Registration is already cancelled"],
    "timestamp": "2026-04-24T19:30:00"
}
```

- **Rules and Validations:**
  - Valid JWT token required in Authorization header
  - Registration must belong to the authenticated customer
  - Registration must exist in the database
  - Registration status is updated to CANCEL
  - Cancellation is only allowed before the seminar start time; attempting to cancel after the seminar has started returns HTTP 400 with message: `"Cannot cancel a registration after the seminar has started"`
  - When a SUCCESS registration is cancelled, the system automatically promotes eligible WAIT registrations to SUCCESS in FIFO order (earliest registration first), and sends a promotion email to each promoted customer
  - updatedAt timestamp is updated to current time

---

10. Get Registration History

Retrieve all seminar registrations for the authenticated customer.

- **Url [GET]:** `/api/registrations/history`
- **Input Parameters:**
  - **Header:**
    - Accept: `application/json`
    - Authorization: `Bearer {JWT_TOKEN}` (Required)
  - **Path Variables / Query Parameters:**
    - None
  - **Body:**
    - None
- **Output in JSON format:**

Response Sample (Success)

```
[
    {
        "id": 101,
        "seminarId": 1,
        "vehicleModelNumber": "EV-2024-001",
```

```
        "seminarDate": "2026-05-15T14:00:00",
        "seatsBooked": 2,
        "status": "SUCCESS",
        "createdAt": "2026-04-24T19:30:00",
        "updatedAt": "2026-04-24T19:30:00"
    },
    {
        "id": 102,
        "seminarId": 3,
        "vehicleModelNumber": "EV-2024-003",
        "seminarDate": "2026-06-01T15:00:00",
        "seatsBooked": 1,
        "status": "WAIT",
        "createdAt": "2026-04-23T10:15:00",
        "updatedAt": "2026-04-23T10:15:00"
    },
    {
        "id": 100,
        "seminarId": 2,
        "vehicleModelNumber": "EV-2024-002",
        "seminarDate": "2026-04-10T14:00:00",
        "seatsBooked": 2,
        "status": "CANCEL",
        "createdAt": "2026-04-05T09:00:00",
        "updatedAt": "2026-04-08T11:30:00"
    }
]
```

Response Sample (Success - No Registrations)

```
[]
```

Response Sample (Fail - Unauthorized)

```
{
    "status": 401,
    "message": "Unauthorized access",
    "errors": ["Unauthorized access"],
    "timestamp": "2026-04-24T19:30:00"
}
```

- **Rules and Validations:**
  - Valid JWT token required in Authorization header
  - Returns all registrations (SUCCESS, WAIT, CANCEL) for the authenticated customer
  - Empty array returned if customer has no registrations
  - Only returns registrations where the seminar date is within the past 1 year from the current date
  - Results are ordered by seminar date descending (most recent first)

11. Get Registration by ID

Retrieve detailed information about a specific registration for the authenticated customer.

- **Url [GET]:** `/api/registrations/{id}`
- **Input Parameters:**
  - **Header:**
    - Accept: `application/json`
    - Authorization: `Bearer {JWT_TOKEN}` (Required)
  - **Path Variables / Query Parameters:**
    - `id`: Registration ID (Long, Required)
  - **Body:**
    - None
- **Output in JSON format:**

Response Sample (Success)

```
{
    "id": 101,
    "seminarId": 1,
    "vehicleModelNumber": "EV-2024-001",
    "seminarDate": "2026-05-15T14:00:00",
    "seatsBooked": 2,
    "status": "SUCCESS",
    "createdAt": "2026-04-24T19:30:00",
    "updatedAt": "2026-04-24T19:30:00"
}
```

Response Sample (Fail - Registration Not Found)

```
{
    "status": 404,
    "message": "Registration not found with id: 999",
    "errors": ["Registration not found with id: 999"],
    "timestamp": "2026-04-24T19:30:00"
}
```

Response Sample (Fail - Unauthorized)

```
{
    "status": 401,
    "message": "You are not authorized to view this registration",
    "errors": ["You are not authorized to view this registration"],
```

```
    "timestamp": "2026-04-24T19:30:00"
}
```

- **Rules and Validations:**
  - Valid JWT token required in Authorization header
  - Registration must belong to the authenticated customer
  - Registration must exist in the database
  - Returns 404 if registration not found
  - Returns 401 if registration belongs to a different customer

---

Common Error Responses

All endpoints may return the following common error responses:

400 Bad Request

Returned when request validation fails or business logic constraints are violated.

```
{
    "status": 400,
    "message": "Validation failed",
    "errors": [
        "field: error message"
    ],
    "timestamp": "2026-04-24T19:30:00"
}
```

401 Unauthorized

Returned when authentication fails or token is invalid/expired.

```
{
    "status": 401,
    "message": "Unauthorized access",
    "errors": ["Unauthorized access"],
    "timestamp": "2026-04-24T19:30:00"
}
```

404 Not Found

Returned when requested resource does not exist.

```
{
    "status": 404,
    "message": "Resource not found",
```

```
    "errors": ["Resource not found"],
    "timestamp": "2026-04-24T19:30:00"
}
```

500 Internal Server Error

Returned when an unexpected server error occurs.

```
{
    "status": 500,
    "message": "An unexpected error occurred",
    "errors": ["An unexpected error occurred"],
    "timestamp": "2026-04-24T19:30:00"
}
```

Authentication Flow

1. **Registration**: User registers via `/api/auth/register` → Receives OTP via email
2. **Verification**: User verifies email via `/api/auth/verify-email` with OTP → Receives JWT token
3. **Login**: Verified user logs in via `/api/auth/login` → Receives JWT token
4. **Protected Endpoints**: Include JWT token in Authorization header as `Bearer {token}`

Data Type Reference

CustomerType Enum

- `COMPANY`: Corporate customer account
- `PERSONAL`: Individual customer account

RegistrationStatus Enum

- `SUCCESS`: Registration confirmed with allocated seats
- `WAIT`: Registration waitlisted due to insufficient seats
- `CANCEL`: Registration cancelled by customer